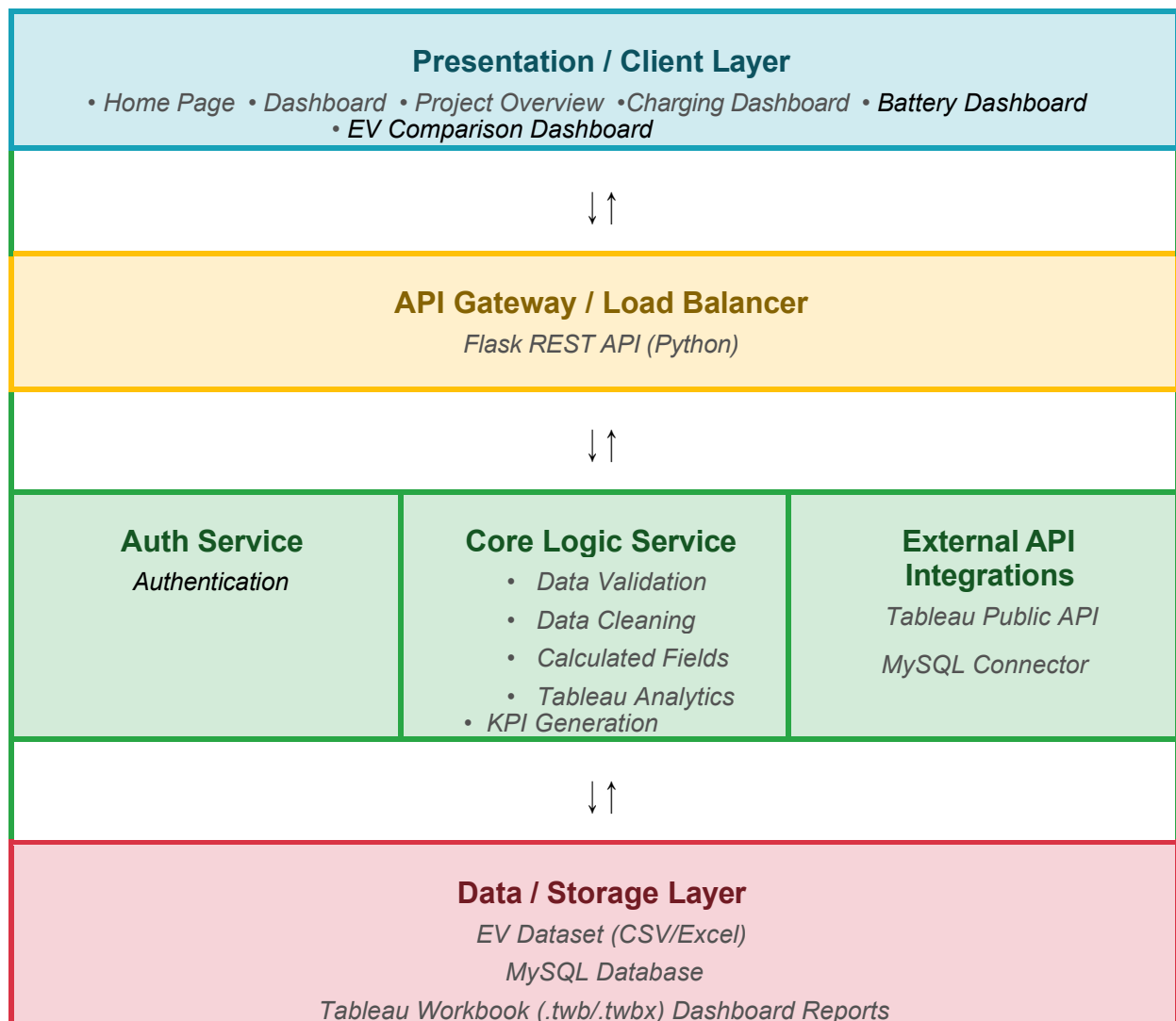


Solution Architecture

Date	10 july 2026
Team ID	XXXXXX
Project Name	EV Data Analytics & Visualization Dashboard
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users upload EV datasets, view interactive dashboards, apply filters, and analyze charging patterns, battery performance, and EV model efficiency.	HTML, CSS, Bootstrap, Tableau Embedded Dashboard
API Gateway	Handles requests from the frontend, validates uploaded datasets, connects the MySQL database, and serves Tableau dashboards through the Flask application.	Flask REST API
Core Logic Service	Validates, cleans, preprocesses, and transforms EV charging, battery, and vehicle data. Performs calculations and prepares data for Tableau visualizations and KPI generation.	Python, Scikit-learn, Pandas, NumPy
Database / Data Layer	<i>Stores EV datasets, charging records, battery information, vehicle details, Tableau workbooks, and dashboard reports for visualization and analysis.</i>	<i>MySQL, CSV Dataset, Tableau Workbook (.twb/.twbx)</i>